Horizontal launches

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Find flight time from vertical motion and then horizontal range.

SUCCESS CRITERIA - I can...

1. Separate vertical and horizontal givens.
2. Find or use fall time before calculating range.
3. Check units and interpret the landing point.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “horizontal” and “vertical” to explain your model or result. Sentence starters are optional.

horizontal: along a level line, usually the x direction

vertical: up and down, usually the y direction

displacement: change from initial position to final position, with direction

01 Warm-up: bring back an earlier idea

At what altitude h/R is a spherical planet’s field 1/9 of its surface value? Repeat for 1/25.

Word help: altitude = height above a reference surface

Sentence starter: My design shows … . It meets the requirement by … .

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $h = \frac{1}{2}gt^2$; range = $v_x t$.
Horizontal launch; h is the positive drop, t is fall time; g = 10 m/s². CPA may use the supplied time.

Worked reasoning: A horizontal launch falls 20 m with g=10 m/s². From $20=5\,t^2$, t=2 s. At horizontal speed 3 m/s, range=6 m.

Your turn: Draw a diagram labeling height and range. Number the two solution steps to show why time comes first.

Sentence starter: Under the assumption …, the evidence … supports … because … .



Add velocity / acceleration arrows.

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

On the diagram, label height h, horizontal range and initial horizontal velocity. Which direction determines the fall time?

Sentence starter: I found ... [unit]. This value means ... in this situation.



Add velocity / acceleration arrows.

Marzano check - circle: 1 2 3

04 Apply the physics

In a launch simulation, a ball leaves a 45 m cliff horizontally at 15 m/s.
Label height and range on the model.
Find the flight time first, then the range.
Use g = 10 m/s² and ignore air resistance.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Add velocity / acceleration arrows.

Marzano check - circle: 1 2 3

05 Explain your reasoning

Two platforms have heights 5 and 20 m. Supplied fall times are 1 and 2 s. At the same launch speed, does quadrupling height quadruple range?
Explain.

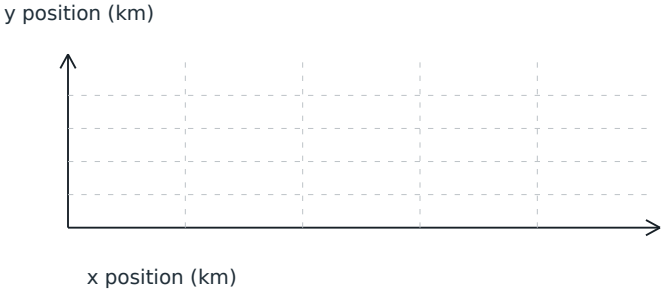
Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A simplified model predicts position (-60, 80) km with bounds ±6 km in x and ±8 km in y. Write a short briefing giving both intervals and two reasons it is not a real asteroid collision forecast.
Label the given and predicted quantities on the model.

Sentence starter: I predict ... . My model supports this because ... .



Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

From a 20 m cliff, horizontal launch speed is 8 m/s.
Find t, range and final vertical velocity.
Label the model with the known quantities before calculating. Ignore air resistance; use g = 10 m/s² when needed.

Sentence starter: I found … [unit]. This value means … in this situation.

Separate horizontal and vertical motion

Label x and y; state positive directions.

Velocity:

Acceleration / force:

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A horizontal launch at 12 m/s must land 18 m from the base.
Find launch height and impact speed using g=10 m/s².
Label the model with the known quantities before calculating. Ignore air resistance; use g = 10 m/s² when needed.

Sentence starter: I found … [unit]. This value means … in this situation.



Add velocity / acceleration arrows.

Marzano check - circle: 1 2 3